

GUÍA 5

5.1

URNA

4V

EXTRAIGO 3

LO PUEDO HACER
CON LA HIPERGEOM.

3A

3R

$$P(x,y) = \frac{\binom{4}{x} \binom{3}{y} \binom{3}{3-x-y}}{\binom{10}{3}}$$

X "CANT BOLAS VERDES"

Y "CANT BOLAS ROJAS"

$$P_{Y|X=X_0} = \frac{P_{XY}(X_0, Y)}{P_X(X_0)}$$

x/y	0	1	2	3	$P_X(x)$
0	$\frac{1}{120}$	$\frac{3}{40}$	$\frac{3}{40}$	$\frac{1}{120}$	$\frac{1}{6}$
1	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{1}{10}$	0	$\frac{3}{10}$
2	$\frac{3}{20}$	$\frac{3}{20}$	0	0	$\frac{6}{20}$
3	$\frac{1}{30}$	0	0	0	$\frac{1}{30}$
$P_Y(y)$	$\frac{7}{24}$	$\frac{21}{40}$	$\frac{7}{40}$	$\frac{1}{120}$	1

$$P(x,y) = \frac{\binom{4}{x} \binom{3}{y} \binom{3}{3-x-y}}{\binom{10}{3}}$$

• $P_X(x)$ = PUNTUAL, SALE DE TABLA

5.2

(X, Y) VECTOR ALEATORIO CONTINUO

HALLAR $f_{Y|X=X_0}(x, y)$, $f_X(x)$

A-

$$(X, Y) \sim U(x^2 + y^2 = 1)$$

COMO ES UNIFORME

$$f_{XY}(x, y) = \frac{1}{\pi 1^2} = \frac{1}{\pi}$$

$$f_X(x) = \int \frac{1}{\pi} dy$$

$$f_X(x) = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi} dy = \frac{1}{\pi} y \Big|_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}$$

$$f_X(x) = \frac{1}{\pi} \cdot 2\sqrt{1-x^2}, \quad \mathbb{I} \mid -1 < x < 1$$

ENTONCES

$$f_{Y|X=X} = \frac{f_{XY}(x, y)}{f_X(x)} = \frac{1}{\pi} \cdot \frac{\pi}{2\sqrt{1-x^2}}$$

$$f_{Y|X=X_0}(y) = \frac{1}{2\sqrt{1-x_0^2}} \mathbb{I} \mid -1 < y < 1$$

B-

$$f_{XY}(x, y) = \frac{1}{2x+1} \cdot e^{-(2x - \frac{y}{4x+2})} \mathbb{I} \mid x > 0, y > 0$$

$$\frac{1}{2x+1} e^{-2x + \frac{1}{4x+2} y} \rightarrow \text{POR PROPIEDAD } e^{A+B} = e^A \cdot e^B$$

TIENE LA PINTA DE DOS EXPONENCIALES

NOTA

$$f_x(x) = 2e^{-2x} \quad x \sim \text{EXP}(2)$$

$$f_{y|x=x} = \frac{1}{4x+2} e^{-\frac{1}{4x+2}} \sim \text{EXP}\left(\frac{1}{4x+2}\right)$$

SI LA MULTIPLICO, ME QUEDA LA CONJUNTA

$$2e^{-2x} \cdot \frac{1}{4x+2} e^{-\frac{1}{4x+2}}$$

$$\frac{2}{2(2x+1)} e^{-(2x + \frac{1}{4x+2})}$$

C-

$$f_{xy}(x,y) = e^{-x} \mathbb{1}_{\{0 < y < x\}}$$

$$f_{y|x=x} = \frac{1}{x} \sim u(0 < y < x)$$

$$f_x(x) = x e^{-x} \sim \text{EXP}(x) \mathbb{1}_{\{x > 0\}}$$

D-

$$f_{xy}(x,y) = \frac{1}{6} x^4 y^3 \cdot e^{-xy} \mathbb{1}_{\{1 < x < 2, y > 0\}}$$

$$f_{xy}(x,y) = \frac{1}{6} x^3 y^3 \cdot x e^{-xy}$$

$$\frac{1}{6} x^4 \cdot y^3 \cdot e^{-xy}$$

$$\frac{\lambda^x}{\Gamma(x)} y^{x-1} \cdot e^{-\lambda y}$$

$$\lambda = x$$

$$y^3 = y^{x-1} \rightarrow x = 4$$

$$x^4 = \checkmark$$

$$\Gamma(4) = 3! = 6$$

$$f_{Y|X=X} = \frac{\lambda^x}{\Gamma(x)} y^{x-1} e^{-\lambda y}$$

$$f_{Y|X=X} = \frac{x^4}{6} y^3 \cdot e^{-xy} \cdot \mathbb{1}_{\{y>0\}}$$

$$F_X(x) = 1 \sim \mathcal{U}(1,2)$$

$$F_X(x) = \frac{1}{1} \mathbb{1}_{\{1 < x < 2\}}$$

5.3

Y: "MILES DE LITROS DE LECHE"

X: "N.º LECHE VENDIDA"

A- $f_{X,Y} = 0,5 \mathbb{1}_{\{0 < x < y < 2\}}$

$$F_X(x) = \int_x^2 0,5 dy = 0,5 y \Big|_x^2 = 0,5(2-x)$$

$$F_X(x) = 1 - \frac{1}{2}x \mathbb{1}_{\{0 < x < 2\}}$$

$$f_{Y|X=1,5}(y) = \frac{0,5}{1 - \frac{1}{2}x} = \frac{0,5}{1 - \frac{1}{2} \cdot 1,5} = 2 \mathbb{1}_{\{x < y < 2\}}$$

$$f_{X|Y=0,8}(x) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$f_Y(y) = \int_0^y 0,5 dx = 0,5y$$

$$f_{X|Y=0,8}(x) = \frac{0,5}{0,5y} = \frac{0,5}{0,5 \cdot 0,8} = 1,25 \mathbb{1}_{\{0 < x < y\}}$$

NOTA

B-

$$P(1,75 < Y < 2 \mid X = 1,5)$$

$$P = \frac{P(1,75 < Y < 2 \cap X = 1,5)}{P(X = 1,5)}$$

COMO YA TENGO LA MARGINAL CALCULO DIRECTO

$$\cdot P(1,75 < Y < 2 \cap X = 1,5) = \int_{1,75}^2 2 \mathbb{I}\{4,5 < Y < 2\} = 0,5$$

$$P(0,5 < X < 0,75 \mid Y = 0,8) = \int_{0,5}^{0,75} \frac{5}{4} \mathbb{I}\{0 < X < 0,8\} = 0,3125$$

C- X e Y SON INDEP.?

$$f_{XY} = f_X \cdot f_Y$$

$$0,5 \neq (1 - \frac{1}{2}X)(0,5Y) \text{ NO SON INDEPENDIENTES}$$

5.4

$$f_{XY} = \frac{5}{8\pi} e^{-\frac{25}{32}(x^2 - 6xy + y^2)}$$

COMPLETO CUADRADOS PARA BUSCAR LLEVAR A UNA NORMAL

$$y^2 + x^2 - \frac{6}{5}xy = \underbrace{y^2 - 2\left(\frac{3}{5}x\right)y + \left(\frac{3}{5}x\right)^2}_{\left(y - \frac{3}{5}x\right)^2} - \underbrace{\left(\frac{3}{5}x\right)^2}_{\frac{16x^2}{25}} + x^2$$

$$\begin{aligned} f_{XY}(x,y) &= \frac{5}{8\pi} e^{-\frac{25}{32}\left[\left(y - \frac{3}{5}x\right)^2 + \frac{16}{25}x^2\right]} \\ &= \frac{5}{8\pi} e^{-\frac{25}{32}\left(y - \frac{3}{5}x\right)^2} e^{-\frac{25 \cdot 16}{32 \cdot 25}x^2} \\ &= \frac{5}{8\pi} e^{-\frac{1}{2}\left(y - \frac{3}{5}x\right)^2} e^{-\frac{1}{2}x^2} \\ &= \left(\frac{1}{\sqrt{2\pi} \cdot 4/5} e^{-\frac{1}{2}\left(\frac{y - \frac{3}{5}x}{4/5}\right)^2}\right) \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}\right) \cdot \frac{1}{5 \cdot 8\pi} \end{aligned}$$

SI NECESITABA

$$f_{X|Y=Y}$$

$$\leftarrow f_{Y|X=X}$$

$$f_X(x)$$

COMPLETO CUADRADOS PARA X.

$$\begin{aligned} P(1 < XY < 5 \mid X = \sqrt{5}) &= P\left(\frac{1}{\sqrt{5}} < Y < \frac{5}{\sqrt{5}} \mid X = \sqrt{5}\right) \\ &= P\left(\frac{1}{\sqrt{5}} < Y \mid X = \sqrt{5} < \frac{5}{\sqrt{5}}\right) \end{aligned}$$

5.5

$$X \sim U(3, 4)$$

$$f_X(x) = 1 \quad \text{if } 3 < x < 4$$

$$f_{Y|X=X} \sim N(X, 1)$$

$$x \in (3, 4)$$

A=

CALCULAR $f_Y(5)$

$$f_{XY} = f_X(x) \cdot f_{Y|X=x}$$

$$f_{XY} = \frac{1}{\sqrt{2\pi}x} e^{-\frac{(y-x)^2}{2}} \mathbb{I}\{3 < x < 4\}$$

$$f_Y(y) = \int_0^{+\infty} f_{XY} dx = \int_3^4 \frac{1}{\sqrt{2\pi}} e^{-\frac{(y-x)^2}{2}}$$

$$\downarrow x \in (3, 4)$$

$$f_Y(5) = \int_3^4 \frac{1}{\sqrt{2\pi}} e^{-\frac{(5-x)^2}{2}} = P(3 < W < 4) = -$$

$$W \sim N(5, 1)$$

$$\Phi\left(\frac{3-5}{1}\right) - \Phi\left(\frac{4-5}{1}\right)$$

$$B \quad P(X > 3.5 | Y = 5) = P(X | Y = 5 > 3.5)$$

$$f_{X|Y=y_0} = \frac{f_{XY}(x, y_0)}{f_Y(y_0)}$$

$$f_{X|Y=5} = \frac{1}{\sqrt{2\pi}x} e^{-\frac{(5-x)^2}{2}} \cdot \frac{1}{0.136}$$

$$P(X | Y = 5 > 3.5) = 1 - \int_0^{3.5} \dots =$$

$$\begin{aligned} & 1 - 0.136 \\ & 0.02375 - 0.159 \\ & \Phi(-2) - \Phi(-1) \\ & \downarrow \\ & \Phi(-z) = 1 - \Phi(z) \end{aligned}$$

5.7

$$T_1 \sim \text{EXP}(\lambda)$$

$$T_2 \sim \text{EXP}(\lambda)$$

 T_1 Y T_2 SON INDEPENDIENTES

$$S_1 = T_1 \rightarrow S_1 \sim \text{EXP}(\lambda)$$

$$S_2 = T_1 + T_2$$

A- HALLAR f_{S_1, S_2}

$$f_{T_1 T_2} = f_{T_1} \cdot f_{T_2}$$

$$= \lambda e^{-\lambda T_1} \cdot \lambda e^{-\lambda T_2}$$

$$= \lambda^2 e^{-\lambda(T_1 + T_2)}$$

$$(S_1, S_2) = g(T_1, T_2) = (T_1, T_1 + T_2)$$

$$f_{S_1 S_2}(s_1, s_2) = \frac{f_{T_1 T_2}(t_1, t_2)}{\left| \det \frac{\partial g(t_1, t_2)}{\partial (t_1, t_2)} \right|} \quad (t_1, t_2) = g^{-1}(s_1, s_2)$$

$$\begin{cases} S_1 = T_1 \\ S_2 = T_1 + T_2 \end{cases} \Rightarrow \begin{cases} T_1 = S_1 \\ T_2 = S_2 - S_1 \end{cases} \quad g^{-1}(s_1, s_2) = (s_1, s_2 - s_1)$$

$$f_{S_1 S_2}(s_1, s_2) = \frac{f_{T_1 T_2}(t_1, t_2)}{g^{-1}(s_1, s_2)} \cdot \left| \det \frac{\partial g^{-1}(s_1, s_2)}{\partial (s_1, s_2)} \right|$$

5.8

DRACULA $0:00 + X$
 BANFIELD $0:00 + Y$ } SON INDEP

$$P_{XY} = \frac{1}{35} e^{-\left(\frac{x}{5} + \frac{y}{7}\right)}$$

$$P_X(x) = \frac{1}{5} e^{-\frac{x}{5}} \sim \text{EXP}(1/5)$$

$$P_Y(y) = \frac{1}{7} e^{-\frac{y}{7}} \sim \text{EXP}(1/7)$$

PRIMERO EN LLEGAR TUVO QUE ESPERAR 5 MIN

LO QUE ME PIDEN ES COMO 4.14 (B)

$$J = 1 \text{ if } U = X \text{ } + 2 \text{ if } U = Y$$

↓
PRIMERO
DRACULA

↓
PRIMERO
BANFIELD

$W = \text{TIEMPO DE ESPERA} \rightarrow (W=5)$
 $= \text{MAX} - \text{MIN}$

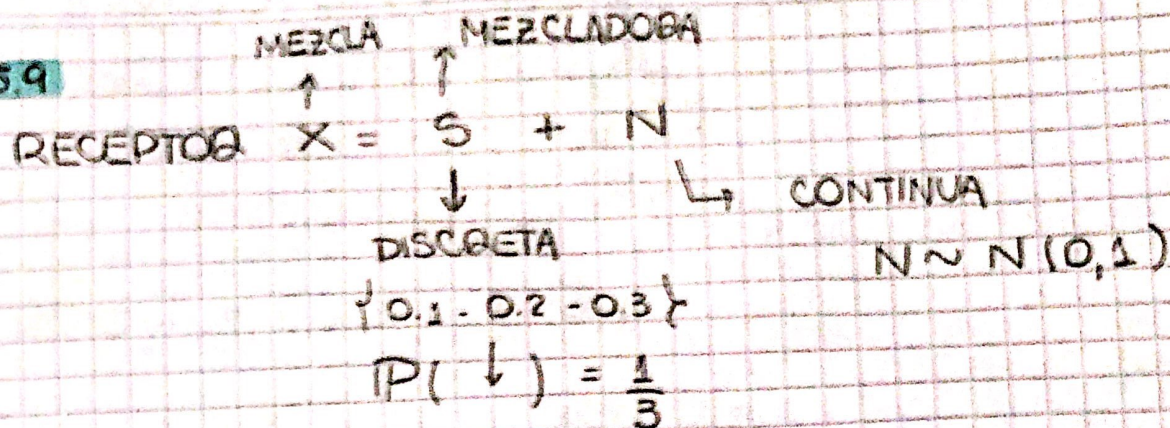
LO TENGO

$$P(J=1 | W=5) = \frac{\int_0^5 f_{W|J=1}(s) \cdot P(J=1) ds}{\int_0^5 f_{W|J=1}(s) \cdot P(J=1) ds + \int_0^5 f_{W|J=2}(s) \cdot P(J=2) ds}$$

↓
BAYES PARA MEZCLAS

$$\int_0^5 f_{W|J=1}(s) \cdot P(J=1) ds + \int_0^5 f_{W|J=2}(s) \cdot P(J=2) ds$$

59



CALCULAR $P(S = 0.2 | X = 0.8?)$

↓ NO PUEDO RESOLVER

XG, NO PUEDO EVALUAR UNA CONTINUA EN UN PUNTO

BAYES PARA MEZCLA DE CONTINUAS, f_x EN CADA PUNTO

$$P(M=i | X_M = x) = \frac{f_{X_i}(x) \cdot P(M=i)}{\sum_{j=1}^m f_{X_j} \cdot P(M=j)}$$

$$X_1 = 0.1 + N$$

$$X_2 = 0.2 + N$$

$$X_3 = 0.3 + N$$

NECESITO SUS $f_x = f_{X_1} f_{X_2} f_{X_3}$

↓ COMO SON UN CAMBIO DE VARIABLE LINEAL $y = Ax + B$

$$X_i = 1N + S_i$$

$\sim N(0,1)$
↑ TABLA.

USO

$$f_{X_i} = \frac{f_x \left(\frac{x-B}{A} \right)}{|A|}$$

$$A=1$$

$$B=S_i$$

$$f_{X_i} = f_x(x - S_i)$$

$$P(M=i) = 1/3 \text{ (EQUIPROBABLE)}$$

$$P(S=0,2 | X=0,87) = \frac{f_{X_2}(0,87) \cdot P(S=0,2)}{f_{X_1}(0,87) \cdot P(S=0,1) + f_{X_2}(0,87) \cdot P(S=0,2) + f_{X_3}(0,87) \cdot P(S=0,3)}$$

$$P(S=0,2 | X=0,87) = \frac{0,749 \cdot 1/3}{0,761 \cdot 1/3 + 0,749 \cdot 1/3 + 0,716 \cdot 1/3} = 0,336$$

5.10

CORPORACION $\rightarrow$ 6 ROBOTS

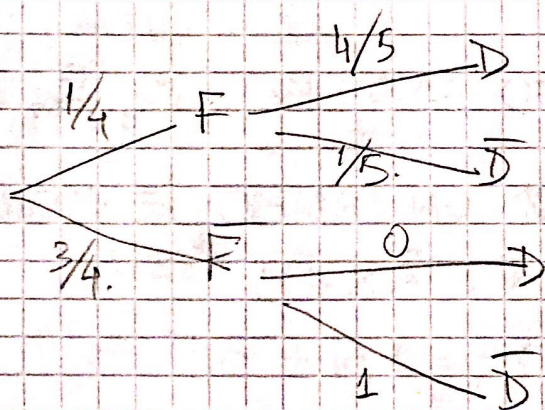
$\downarrow$ PROBABILIDAD DE (C/U)
FALLA = $1/4$

PRUEBA $\rightarrow$ SI FALLA $\rightarrow 4/5$

X : # ROBOTS FALLADOS $\sim Bi(6, 1/4)$

Y : # DETECTADA DE ROBOTS FALLADOS

A- $p(x) = E[Y | X = x_0]$



$$Y | X = x_0 \sim Bi(x_0, \frac{4}{5}) \rightarrow E[Y | X = x_0] = \frac{4}{5} x_0$$

$$X | Y = y \sim y + M \rightarrow E[X | Y = y_0] = 6 - y \cdot P(F|D)$$

$\downarrow$
 $\sim Bin(6-y, P(F|D))$

5.12

ARROJO UN DADO 36 VECES

 X : # PARES Y : # IMPARES $\sim \text{BER}()$ HALLAR $E[Y|X]$ y $\text{COV}(X, Y)$

$$X + Y = 36$$

$$X \sim \text{BIN}(36, P_X) = \text{BIN}(36, 1/2)$$

$$Y \sim \text{BIN}(36, P_Y) = \text{BIN}(36, 1/2)$$

$$X + Y = 36 \quad \begin{cases} X = 36 - Y \\ Y = 36 - X \end{cases}$$

$$E[Y|X] = E[36 - X | X] = 36 - X$$

 $\downarrow$ PROPIEDAD X ESTA FIJO $\rightarrow$ QUEDA CTE